skaraz data



Stéphane Karasiewicz

Independent calculation agent for parametric insurance · Principal, Skaraz Data

Hauts-de-France · Remote worldwide · native
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Principal of Skaraz Data, independent calculation agent: I define, build and independently check the index a parametric policy pays on, from public satellite and reanalysis data.

Experience

Skaraz Data, Principal (Sep 2023 → Present). Constellation (Jun 2026 → Present)

Remote

Independent calculation agent for parametric insurance. Stéphane Karasiewicz EI, trading as Skaraz Data.

- Trigger audits, index design and feasibility, onboarding of one index (definition, pipeline, backtest, methodology report)
- Pipeline support and independent verification; calculation agent service set up per contract
- Five proofs of concept on public data with limits named: coral heat stress, pluvial flood, cyclone wind, crop drought

Raincoat LLC, Senior Data Scientist (contract)

Feb 2024 → Jan 2026 · Remote

Parametric insurtech (Seed-stage, San Juan, Puerto Rico). Climate risk technology.

- Ran the satellite and reanalysis data layer for every peril except hurricane wind and earthquake, across **10+ countries** in Latin America, the Caribbean and Africa: precipitation and drought maintained in production, **water balance** built from scratch to production, **heat stress** built but not shipped
- Engineered automated data management for **2 TB+ satellite & reanalysis data** (CHIRPS, IMERG, ERA5-Land, Copernicus CDS, WEkEO)
- Delivered a **modular architecture roadmap** as a scoped technical handoff for scaling to new perils, transferred to the team at contract end
- Ran dev/ops solo for production pipelines at end of contract; built **3 country-level operational dashboards** (Precipitation, Water Balance, Drought)
- Tools: Python, xarray, GDAL, Docker, PostgreSQL, scikit-learn

IFREMER, Marine Remote Sensing Researcher

Feb 2018 → Jul 2023 · Boulogne-sur-Mer & Brest, France

French National Institute for Ocean Science. 5+ years in satellite remote sensing for marine ecology.

- Operated production R pipelines processing 500 GB+ satellite & in-situ data; reduced manual effort by 50 hours/month
- Authored 2 CRAN packages (BDAlgo, subniche) with **8,000+ cumulative downloads** and 27+ citations
- 7 peer-reviewed publications on environmental modeling using MODIS, Sentinel-2/3, and reanalysis data
- Mentored 6 researchers and Master's students on R, Git, and reproducible research

Technical stack

Climate data: CHIRPS · IMERG · ERA5-Land · Copernicus CDS · WEkEO · MODIS · Sentinel
Engineering: Python (expert) · xarray · GDAL · Docker · PostgreSQL · Airflow · CI/CD
Languages: Python · R (2 CRAN packages) · SQL · Bash
Methods: Production data pipelines · Geospatial analysis · Time-series · Multivariate stats

Education

PhD Marine Ecology, Université de Lille, France (2014-2017). Phytoplankton evolution. Novel WitOMI statistical framework for ecological niches.

M.Sc. Marine Ecology, Stockholm University (2012-2014) · **B.Sc. Ecosystem Science**, Open University, England (2010-2012)

Publications & Open Source

CRAN R packages

- subniche: Ecological niche modeling using OMI framework · 5,000+ downloads · 15+ citations
- BDAlgo: Bloom detection in time-series, extracts 22 phenological variables · 3,000+ downloads · 12+ citations

Featured publication

- Karasiewicz S., Chapelle A., Bacher C., Soudant D. (2020). Harmful algae niche responses. **Harmful Algae**, 98:101785. https://doi.org/10.1016/j.hal.2020.101785

6 additional peer-reviewed publications at ResearchGate.

Languages

French · Native. English · Native (dual citizenship).

Contact

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